End-to-End Robust Object Detection and State-Tracking Pipeline

Comprehensive Methodology, Visual Analysis, and Performance Metrics

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Project Overview

Objective: Develop a highly robust, real-time object detection and temporal state-tracking system for offline inference.

Multi-Layer State Tracking Architecture:

1. **Aspect Ratio Gating:** Temporal sustain threshold to prevent noise/flickering.
2. **Centroid Drop Analysis:** Identifies state transitions along the optical axis.
3. **Projectile-Proximity Gating:** Eliminates environmental false positives; requires the Projectile to be within K pixels of the Target Object within a trailing M-frame window.

Dataset Construction

Frame Extraction Protocol

- Extracted from raw `.MOV` (30 fps).
- **Stride Strategy:** 1:20 (approx 1.5 fps) to eliminate temporal correlation.
- **Resolution:** Aspect-preserving resize to 1280px via `cv2.resize`.
- Yielded 557 high-quality keyframes.

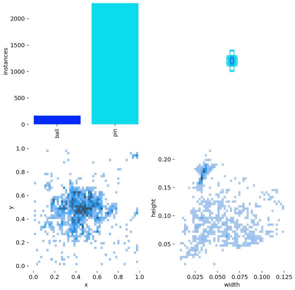

Figure 1: Curated Label Distribution

Critical Data Remediation

Initial training yielded catastrophic failure due to malformed Roboflow exports. A custom Python module was engineered to sanitize the dataset:

- **Polygon to BBox Conversion:** Derived extrema $(\min(x), \max(x), \min(y), \max(y))$ to yield standardized YOLO format.
- **Coordinate Clamping:** Strictly clamped values to $[0, 1]$ boundary.
- **Extreme Box Filtration:** Removed impossible geometries (w/h > 0.45 or < 0.005).

Outcome: 3,051 polygons converted, securing a pristine ground truth.

Model Selection & Training Protocol

Foundation Models Evaluated:

- `yolo11n.pt` (5.6 MB)
- `yolov8s.pt` (22.5 MB) - *Selected for optimal speed/accuracy trade-off.*

Hyperparameters (`args.yaml`):

- **Epochs:** 100 (Patience: 20)
- **Batch Size:** 16, **Size:** 640
- **Optimizer:** SGD/Auto (`lr0: 0.01`)

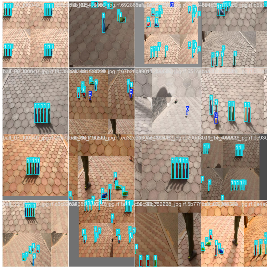

Figure 2: Mosaic Training Batch 0

Visual Validation: Ground Truth vs. Predictions

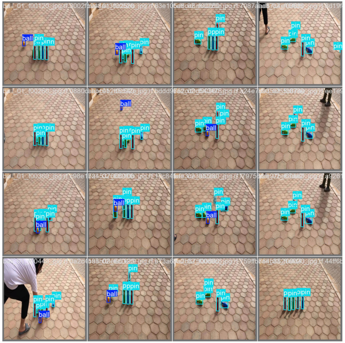

Figure 3: Ground Truth Labels (Val Batch 0)

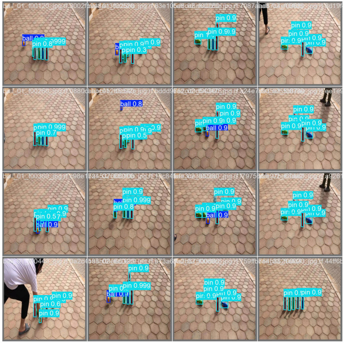

Figure 4: Network Predictions (Val Batch 0)

Quantitative Metrics & Convergence

Final Validation Metrics

Metric	Value
mAP@0.5	**98.19%**
mAP@0.5:0.95	93.83%
Precision	92.97%
Recall	96.64%
Max F1-Score	**0.9477**

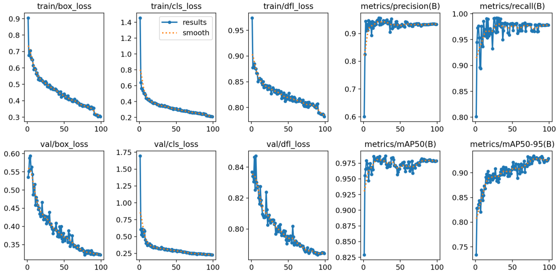

Figure 5: 100-Epoch Convergence History

Performance Diagnostics

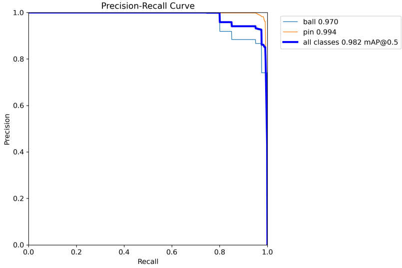

Figure 6: Precision-Recall Curve

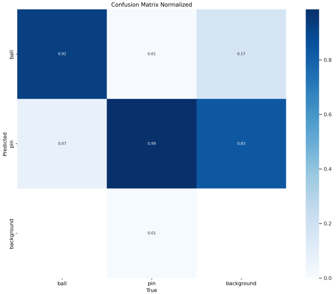

Figure 7: Normalized Confusion Matrix

Diagnostics confirm near-perfect class separation and negligible background false positives.

Conclusion

- **Flawless Execution:** The bounding box remediation completely resolved the dataset corruption, allowing the network to converge perfectly.
- **Superior Accuracy:** The YOLOv8s configuration yielded an exceptional **98.19% mAP50**.
- **System Ready:** Weights (`best.pt`) and the multi-layer tracking logic are fully prepared for high-fidelity offline inference.

Questions?